

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

## Practical - VII

|                              |                                   |
|------------------------------|-----------------------------------|
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| Class :SYIT                  | Batch : 01                        |
| Date of Assignment :05/08/26 | Date/Time of Submission :05/08/26 |

Hash Table:

Write a program to:

- Implement a hash table with separate chaining for collision handling.

Input:

```
size= 5
table=[[ ]for _ in range(size)]
def insert(key, value):
    table[(key%size)].append((key,value))
insert (10,"Apple")
insert (25,"Banana")
insert (15,"Cherry")
insert (20,"Grapes")
insert (45,"watermellon")
insert (32,"Mango")

print (table)
print("S026 Sakshi Patade")
```

Output:

```
>
= RESTART: C:/Users/sejal/OneDrive/Desktop/DATA STRUCTURE/prac7ds.py
[[ (10, 'Apple'), (25, 'Banana'), (15, 'Cherry'), (20, 'Grapes'), (45, 'watermellon') ], [], [(32, 'Mango')], [], []]
S026 Sakshi Patade
>
```

- b. Store and retrieve data from the hash table.

Input:

```
size= 5
table=[[ ]for _ in range(size)]
def insert(key, value):
    table[(key%size)].append((key,value))

def search (key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not Found"

insert (10,"Apple")
insert (25,"Banana")
insert (15,"Cherry")
insert (20,"Grapes")
insert (45,"watermellon")
insert (32,"Mango")

print (table)

key=int(input("Enter Key: "))
print(search(key))
print("S026 Sakshi Patade")
```

Output:

```
== RESTART: C:/Users/sejal/OneDrive/Desktop/DATA STRUCTURE/prac7ds.py
[[ (10, 'Apple'), (25, 'Banana'), (15, 'Cherry'), (20, 'Grapes'), (45, 'watermellon') ], [], [(32, 'Mango')], [], []]
Enter Key: 10
Apple
S026 Sakshi Patade
>>>
== RESTART: C:/Users/sejal/OneDrive/Desktop/DATA STRUCTURE/prac7ds.py
[[ (10, 'Apple'), (25, 'Banana'), (15, 'Cherry'), (20, 'Grapes'), (45, 'watermellon') ], [], [(32, 'Mango')], [], []]
Enter Key: 29
Not Found
S026 Sakshi Patade
```

### Curiosity Questions:

1. In a BST, which side (left or right) has **smaller numbers** than the root?
2. If we insert numbers 10, 20, 5 into a BST, which number will be at the root?
3. What will the inorder traversal of a BST give — sorted numbers or random numbers?
4. Where will the **largest number** in a BST always be found?
5. If the root of a BST is 50 and we search for 70, will we go left or right? Why?
6. If a BST has only **one node**, what is its inorder traversal?
7. Can we create a BST from characters like A, B, C instead of numbers?
8. If we insert numbers 5, 10, 15, 20 in order, will the tree be balanced or like a straight line?
9. In a BST, do we check the left child first or the right child first in inorder traversal?
10. If we delete the root of a BST, will the tree disappear?